

Hackathon Problem Statements

Regd No: 25091A3298, 25091A3297

Problem Statement No: 19

Ambulance Dispatch Route Optimizer Develop an ambulance routing system using Dijkstra's algorithm with dynamically weighted edges representing live traffic and road conditions to minimize response time during emergencies. Integrate historical accident/call data to predict high-demand zones using clustering, enabling smarter ambulance pre-positioning. Model ambulances, hospitals, and incidents as OOP entities with status tracking. This tackles the critical golden-hour delay problem in emergency medical response, especially in tier-2/3 cities.

Regd No: 25091A32D6, 25091A32D3

Problem Statement No: 58

Scholarship Eligibility & Disbursement System Build a scholarship management system using an OOP-based rule validation engine that checks student eligibility against multiple criteria (income, caste category, academic performance, attendance) drawn from various government scheme rules. Use hashing for fast duplicate-application detection across large applicant pools. This reduces fraudulent claims and manual processing delays common in scholarship disbursement schemes.

Regd No: 25091A3217, 25091A3203

Problem Statement No: 63

Exam Seating Arrangement Optimizer Create an exam seating arrangement system using graph coloring/backtracking algorithms to ensure no two students from the same class or with similar roll numbers sit adjacent, minimizing malpractice opportunities across multiple exam halls. Use OOP to model halls, seats, and students with constraint rules. This automates a currently manual and error-prone administrative task for large-scale examinations.

Regd No: 25091A3202, 25091A3249

Problem Statement No: 79

Landfill Capacity Forecasting System Build a landfill capacity forecasting tool using time-series analysis on waste generation trends and current fill-rate data to predict when a landfill will reach capacity, supporting proactive planning for new sites or waste-reduction campaigns. Use OOP to model landfills, waste inflow records, and capacity thresholds. This helps municipal bodies avoid the crisis-mode landfill overflow situations common in many Indian cities.

Regd No: 25091A32G9, 25091A32K4

Problem Statement No: 105

Peer-to-Peer Lending Matchmaking System Build a P2P lending platform backend using graph-based risk-compatibility scoring to match lenders with borrowers based on risk appetite, loan amount, and repayment terms. Use OOP to model lenders, borrowers, loans, and matching logic. This creates an alternative credit access channel for borrowers underserved by traditional banking, while giving lenders more informed matching options.

Regd No: 25091A32N9, 25091A32P9

Problem Statement No: 108

Emergency Resource Dispatch Optimizer Build an emergency resource dispatch system using priority queues combined with graph-based routing to allocate limited resources (rescue teams, medical supplies, food) to disaster-affected areas based on urgency and accessibility. Use OOP to model resources, requests, and dispatch units. This maximizes the impact of limited disaster response resources during the critical early hours of a crisis.